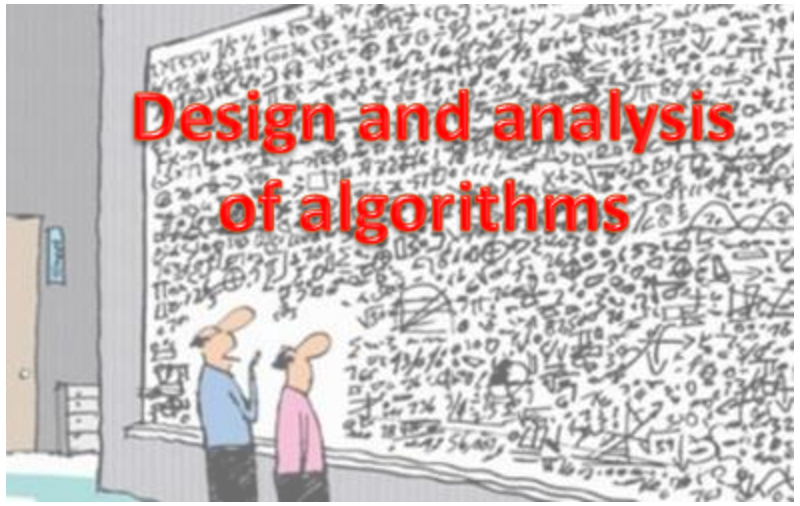


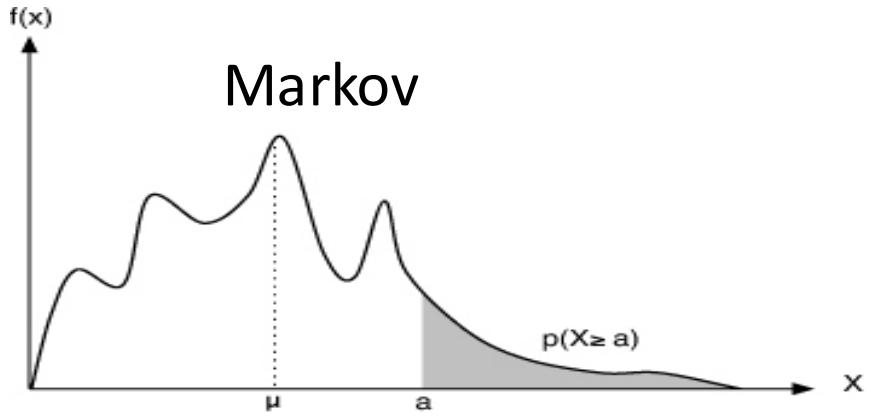
- **Concentration inequalities**
- **Balls and bins**

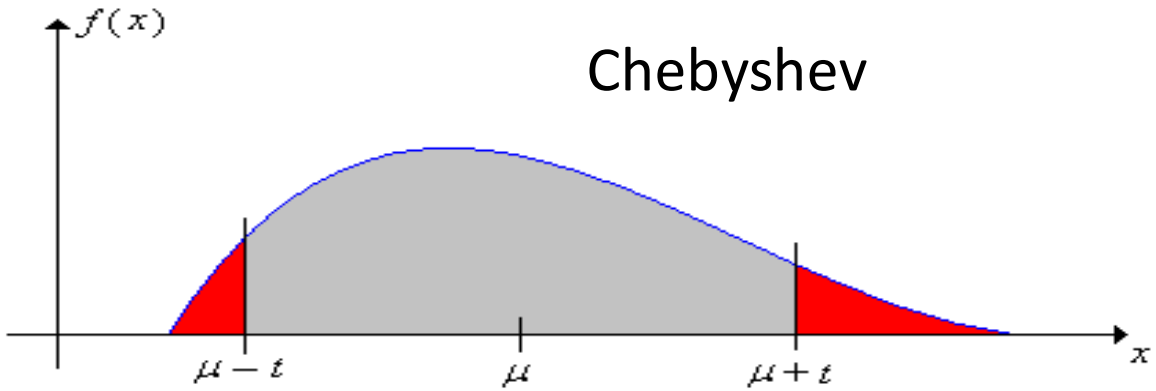
Motivation

- Very general and abstract model: *n* balls, *m* bins.
- Random choice of bin for each ball.
- Study the stochastic properties of the bin occupancy process.
- Applications: hashing, scheduling, …
- First, we introduce some concentration bounds:
  - If we know the mean of a r.v. $E[X]$, how to estimate the probability that X takes a larger value.
  - In many cases we like to show that this can happen very infrequently.
  - There are ways to get upper bounds for this probability. If these are small enough, then we know that the actual probability is also small.

Concentration bounds

- Bound the probability of a random variable to be far from its mean.
- Use very simple information about the random variable.
  - We only know the mean: $\mu = E[X]$: Markov inequality.
  - We know the mean $\mu = E[X]$ and the variance $E[(X-\mu)^2]$: Chebyshev's inequality.
  - If X is the sum of iid bounded variables: Chernoff and Hoeffding's bounds.



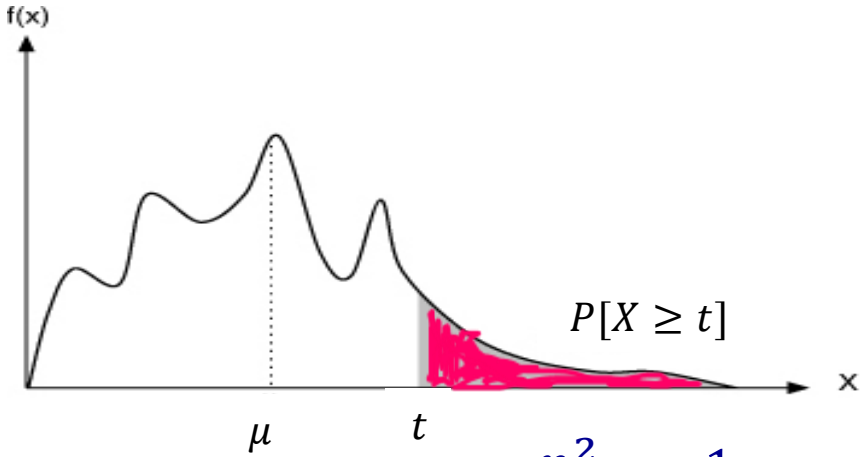


Markov's inequality

- Bounds to probability of an "unlikely" behaviour if we know the average behaviour (i.e., execution time).

$$P[X \geq t] \leq \frac{E[X]}{t}.$$

need X non-negative rv



- Example: $X =$ number of steps to find a solution.

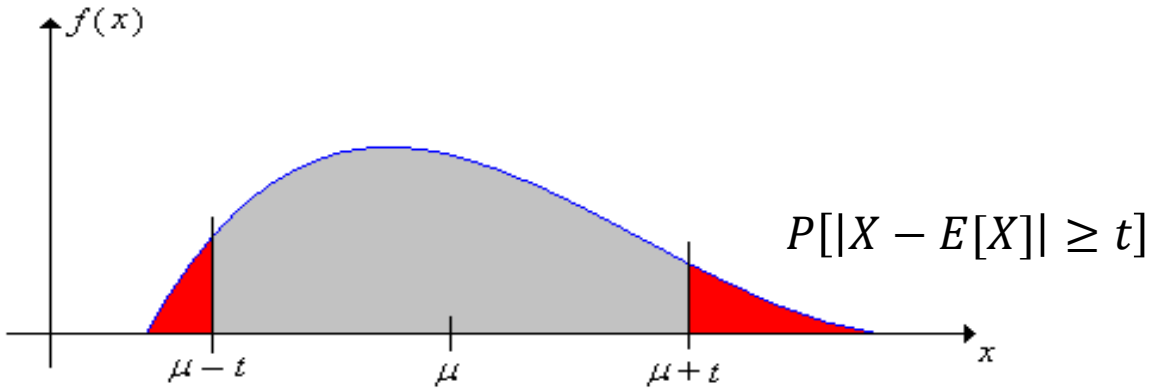
  Suppose we know that $E[X] = n^2$ steps.

  Then the probability that we need more than $2n^2$ steps is $P[X \geq 2n^2] \leq \frac{n^2}{2n^2} = \frac{1}{2}$.

  Similarly, $P[X \geq n^3] \leq \frac{n^2}{n^3} = \frac{1}{n}$.

  Note: Markov's inequality gives a "rough" bound, we may be able the get tighter upper bounds by using other special tools (concentration bounds).

Chebyshev's inequality



$$P[|X - E[X]| \geq t] \leq \frac{Var[X]}{t^2} = \frac{E[(X-\mu)^2]}{t^2}.$$

- To use this, we need to know the variance of X (not just the mean as in the case of Markov's inequality.
- It gives a tighter bound as t increases away from the mean ($1/t^2$ instead of $1/t$).
- When X is a sum of iid rvs, Chernoff and Hoeffding's bounds give exponential e^{-t} bounds (i.e., are more accurate to describe what happens), see slides in appendix .

Hoeffding's inequality

- Let $S_n = x_1 + \cdots + x_n$, where $x_i \in \{0,1\}$ and independent. Then

$$P[|S_n - E[S_n]| > t] < 2e^{-t^2/n}.$$

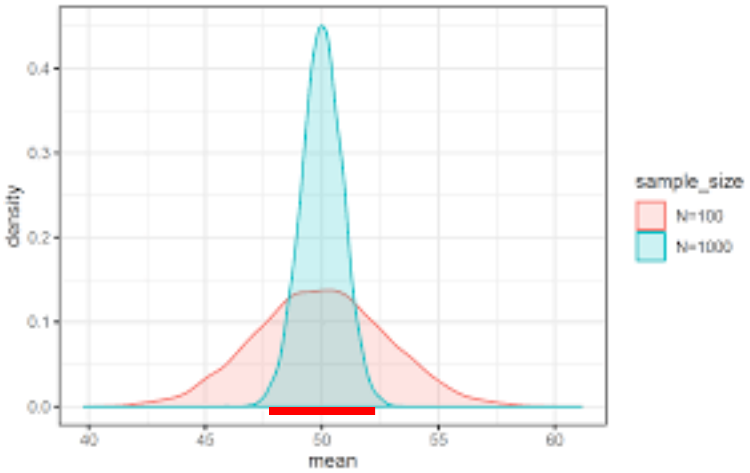
- Equivalently: if $E[x_i] = \mu$, S_n/n = sample mean, $E\left[\frac{S_n}{n}\right] = \mu$,

$$P\left[\left|\frac{S_n}{n} - \mu\right| > \alpha\right] < 2e^{-a^2 n}.$$

use above $t = n\alpha$

- The probability that the sample average $\frac{S_n}{n}$ is outside the disk $\mu \pm \alpha$ goes to zero exponentially fast as n increases!

How probable is the sample average $\frac{x_1+\cdots+x_n}{n}$ to be away from the mean $\mu = E[x_i]$?
we know from the law of large numbers that $\frac{x_1+\cdots+x_n}{n} \to \mu$



Balls and bins

- Very general model: we throw n balls to m bins
- Questions: average number of balls in a bin, average number of empty bins, average number of collisions, how many balls on the average we need to throw to have all bins full, average number of balls in the most loaded bin,…
- Relation to hashing: we assume that the hash function distributes the n keys (balls) randomly over the m locations (bins) of the hash table (SUHA).
- Relation to job scheduling: we have n jobs and m machines and must assign jobs (balls) to these machines (bins). Assume $m = n$
  - Case of **no information**: randomly assign jobs to machines using a hash function. Worse load of a machine $= \log n \,/\, \log\log n$ whp.
  - Case of "**little**" information: The "**power of two choice**" phenomenon: for each job, choose randomly two machines (i.e., use 2 hash functions) and assign the job to the less loaded machine between the two. Then worse load = $\log\log n \,/\, \log\log\log n$ whp.

Simple problems for n balls and m bins

- $E[X_i]$: expected number of balls in bin i.

  $X_{ik} = 1\{\text{ball } k \text{ goes to bin } i\},\ X_i = \sum_k X_{ik}.$

  $E[X_i] = E[\sum_k X_{ik}] = \sum_k E[X_{ik}] = nE[X_{i1}] = \frac{n}{m}.$

$E[1\{A\}]$
$= 1P[A] + 0P[A^c]$
$= 1P[A]$

- $E[Z]$: expected number of empty bins.

  $Z_i = 1\{\text{bin } i \text{ is empty}\},\ Z = \sum_{i=1,\ldots,m} Z_i.$

  $E[Z_i] = P[\text{no ball goes to bin } i] = \left(1 - \frac{1}{m}\right)^n.$

  $E[Z] = E[\sum_i Z_i] = mE[Z_1] = m\left(1 - \frac{1}{m}\right)^n.$

If $m = n$ and n is large, then $E[Z] \approx m\frac{1}{e}$.

The birthday paradox

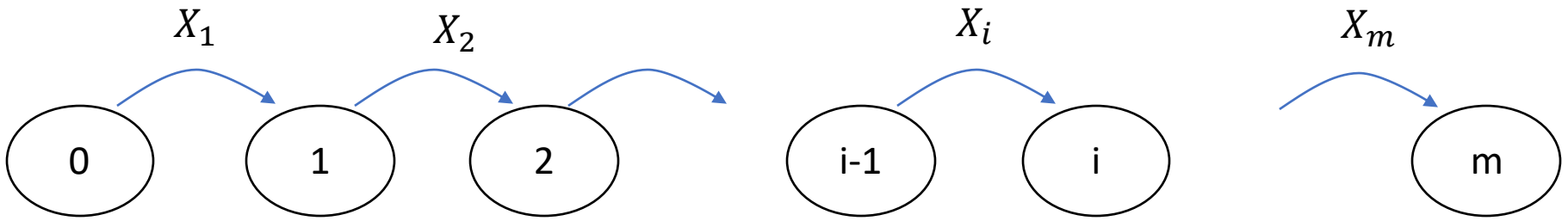
Find the expected number of collisions.

- Collision between i, j where $i \neq j$: $X_{ij} = 1\{\text{balls } i, j \text{ go in the same bin}\}$.
- Number of collisions: $X = \sum_{i<j} X_{ij}$ (we require $i \neq j$ and we count once X_{ij} and X_{ji}).
- Note that $E[X_{ij}] = 1/m$. Hence

$$E[X] = \sum_{i<j} E[X_{ij}] = \frac{1}{m}(\#\text{ of pairs } (i,j) \text{ in the sum}) = \frac{1}{m}\binom{n}{2}.$$

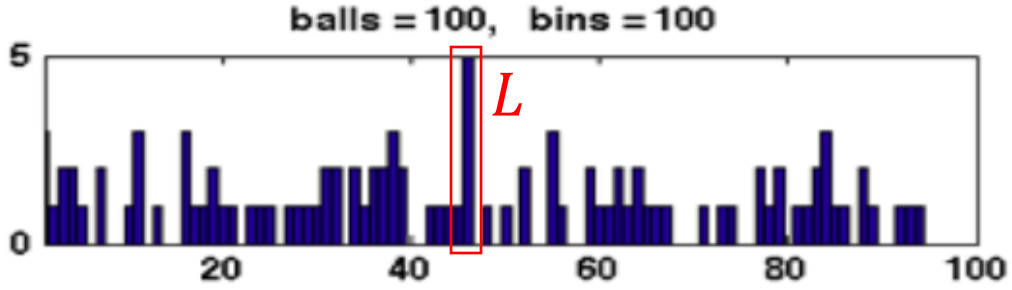
- Birthday question: if balls are people and bins are days of the year, how many people we need to have average number of "birthday collisions" = 1? (i.e., on the average there is one pair of people having bday on the same day).
- Solution: $E[X] = \frac{1}{365}\binom{n}{2} = 1$, solve for n. We need $n = 28$.

Expected time until all bins are full



- X_i = # of balls added between the time when there are $i-1$ full bins until there are i full bins.
- Prob of success (get an empty bin) = $p_i = \frac{m-(i-1)}{m}$.
- Expected number of trials until success = $\frac{1}{p_i} = E[X_i]$.
- Total number of balls X until all m bins have at least a ball: $X = X_1 + X_2 + \cdots + X_m$.
- $E[X] = \sum_{i=1,\ldots,m} E[X_i] = \sum_{i=1,\ldots,m} \frac{m}{m-(i-1)} = m \sum_{j=1,\ldots,m} \frac{1}{j} \approx m \log m$.

The max-load process



- Suppose $n = m$.
- L = max load = max # of balls in any of the bins.
- **Proposition**: $L = O\left(\frac{\ln n}{\ln \ln n}\right)$ **whp**: with probability $> 1 - \frac{1}{n^2}$.
- $L = O\left(\frac{\ln n}{\ln \ln n}\right)$ holds in "almost all" experiments for large n.
- Stronger than $E[L] = O\left(\frac{\ln n}{\ln \ln n}\right)$.

whp: "with very high probability", i.e., with probability that asymptotically tends "fast" to 1, like $1 - \frac{1}{n^c}, c \geq 1$.

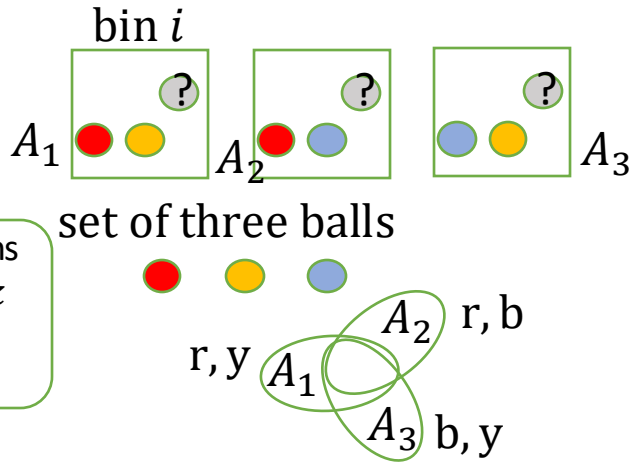
$\ln \ln n$: very slow…

- We break the proof in three lemmas.

Lemma 1: $\Pr\{\text{bin } i \text{ has} \geq k \text{ balls}\} \leq \binom{n}{k}\left(\frac{1}{n}\right)^k$.

union over all possible combinations of k balls

probability that bin i contains a specific combinations of k balls out of the n balls (possibly with other balls)



$\{\text{bin } i \text{ has} \geq 2 \text{ balls}\} = A_1 \cup A_2 \cup A_3$

- Example: set of 3 balls {red, yellow, blue}.
- Event $E = \{\text{bin } i \text{ has} \geq 2 \text{ balls}\}$.

not exclusively

- $E = \{\text{balls red and blue are in bin } i\} \cup \{\text{balls red and yellow are in bin } i\} \cup \{\text{balls yellow and blue are in bin } i\} = A_1 \cup A_2 \cup A_3$.
- This is because every time E is True it implies $A_1 \cup A_2 \cup A_3$, and the reverse also holds, i.e., $A_1 \cup A_2 \cup A_3$ implies E.
- Hence $\Pr[E] = \Pr[A_1 \cup A_2 \cup A_3] \leq \Pr[A_1] + \Pr[A_2] + \Pr[A_3]$ (since not disjoint).

$$= 3\Pr[A_1] = 3\left(\frac{1}{3}\right)^2 = \binom{3}{2}\left(\frac{1}{3}\right)^2.$$

Lemma 2: $\Pr\left\{\text{bin } i \text{ has} \geq \frac{4\ln n}{\ln\ln n} \text{balls}\right\} < \frac{1}{n^3}$.

we use the "death bed" inequality

$$\left(\frac{x}{y}\right)^y \leq \binom{x}{y} \leq \left(\frac{ex}{y}\right)^y$$

Proof:

Step 1: $\Pr\{\text{bin } i \text{ has} \geq k \text{ balls}\} \leq \binom{n}{k}\left(\frac{1}{n}\right)^k \leq \left(\frac{en}{k}\right)^k\left(\frac{1}{n}\right)^k = \left(\frac{e}{k}\right)^k$. (*)

Step 2: choose $k = \frac{4\ln n}{\ln\ln n}$, using (*) we obtain

- $\Pr\left\{\text{bin } i \text{ has} \geq \frac{4\ln n}{\ln\ln n} \text{balls}\right\} \leq \left(\frac{e}{\frac{4\ln n}{\ln\ln n}}\right)^{\frac{4\ln n}{\ln\ln n}}$ % using (*)

$$< \left(\frac{\ln\ln n}{\ln n}\right)^{\frac{4\ln n}{\ln\ln n}} \quad \% \text{ since } \frac{e}{4} < 1$$

$$= \exp\left(\frac{4\ln n}{\ln\ln n}\ln\frac{\ln\ln n}{\ln n}\right) = \exp\left(\frac{4\ln n}{\ln\ln n}(\ln\ln\ln n - \ln\ln n)\right)$$

$$= \exp\left(-4\ln n + 4\ln n\frac{\ln\ln\ln n}{\ln\ln n}\right)$$

$$< \exp(-3\ln n) \quad \% \text{ since } 4\frac{\ln\ln\ln n}{\ln\ln n} \text{ can become arbitrarily small for large } n.$$

$$= \frac{1}{n^3}.$$

Lemma 3: $\Pr\{\text{some bin has} \geq \frac{4\log n}{\log\log n}\text{ balls}\} \leq \frac{1}{n^2}$.

We use the union bound:

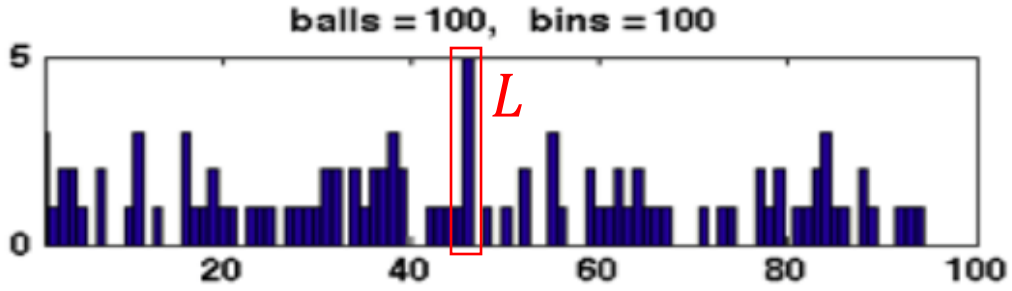
we use union bound since the event is the union of all events for each specific bin

$$\Pr\{\text{some bin has} \geq \frac{4\log n}{\log\log n}\text{ balls}\} \leq \sum_i \Pr\{\text{bin } i \text{ has} \geq \frac{4\log n}{\log\log n}\text{ balls}\} < n\frac{1}{n^3} = \frac{1}{n^2}.$$

- Same as: with prob $> 1 - \frac{1}{n^2}$ no bin has more than $\frac{4\log n}{\log\log n}$ balls,
- Or, with prob $> 1 - \frac{1}{n^2}$ the most loaded bin has $O(\frac{\log n}{\log\log n})$ balls.
- This complete the proof of the Proposition.

- One can prove using similar arguments that the most loaded bin has $\Omega(\frac{\log n}{\log\log n})$ balls with prob $> 1 - \frac{1}{n^2}$.

The max-load process



- Suppose $n = m$
- L = max load = max # of balls in any of the bins
- **Proposition**: $L = O\left(\frac{\ln n}{\ln \ln n}\right)$ **whp**: with probability $> 1 - \frac{1}{n^2}$.
- $L = O\left(\frac{\ln n}{\ln \ln n}\right)$ holds in "almost all" experiments for large n.
- Stronger than $E[L] = O\left(\frac{\ln n}{\ln \ln n}\right)$.
- We can actually prove that whp $L = \Theta\left(\frac{\ln n}{\ln \ln n}\right)$.
- With "choice of two", i.e., "minimum" load balancing:

$$L = \Theta\left(\frac{\ln \ln n}{\ln \ln \ln n}\right)$$

The power of two choices instead of one

- The Proposition says that the most loaded bin is "about" $\frac{\ln n}{\ln \ln n}$.
- Let's suppose that before inserting a ball, we pick two bins at random and chose the less loaded one.
- How is the distribution of the maximum load in this system?
- It drops to $\frac{\ln \ln n}{\ln \ln \ln n}$
- A small amount of information makes a big difference in load balancing!
- Hashing: use 2 hash functions instead of one, choose the slot with the smallest number of collisions.
- Can we do insertions in $O(1)$?

Summary

- Balls that are thrown in bins uniformly at random is a great model to study many practical situations like hashing and job scheduling.
- If $m = n$ or more generally, when the number of bins grows at the same rate as the number of balls, we see that the worse case load is rarely higher than few times $\frac{\ln n}{\ln \ln n}$. This is not an average case result but holds for any experiment!
- If we are allowed to check two random bins before placing a ball, this reduces to $\frac{\ln \ln n}{\ln \ln \ln n}$. Little information about the state of the system can greatly help in balancing system load!